

CHAPTER 13 OF 13 • PART SEVEN

The Map Back

Every entry landed somewhere specific in the feed design, and the most useful one is the entry we refused.



Thirteen chapters, gathered into six movements.

01 A Tuesday — one engineer, wake to sleep

02 The words — naming what you just watched

03 Protagonist and antagonist — where a design starts

04 Solution types — which answer is wanted

05 Six kits — data, compute, network, scale, reliability, security

→ **Two designs, then the map back — Instagram, a parking app, and what you keep**

WHERE THIS SITS

This chapter reads the two designs back against the ten before them.

It assumes chapter eleven's feed design in detail and the six kits from chapters five to ten. Chapter four's removal test runs on the finished design, six review questions run on the parking app from memory, and the deck closes on the two movements it did not teach — develop and deliver — which are the only real check on the two it did.

01

PART SEVEN

Now map the feed design
back to the kits, including
the entry we refused.

Every store-and-run entry landed somewhere specific in the feed design.

KIT ENTRY	WHERE IT LANDED	WHAT IT CHARGES YOU
Wide-column	The two edge tables	The primary key is the schema. A new question means rewriting the data
Object store	Photo bytes and variants	Listing is slow, so the index of what you stored lives somewhere else
Key-value	The feed cache	No second way in. Every new question is a new key you maintain
Durable log	Post events into fan-out	At-least-once. Every consumer here has to be idempotent
Stateless services	Feed and post service	Nothing may be remembered in the process, so state moves and costs a hop
Bounded queue	Fan-out and transcode	A bound means shedding. When it fills, something waits or is dropped

The network, scale, reliability and security kits landed here too.

KIT ENTRY	WHERE IT LANDED	WHAT IT CHARGES YOU
Reduce	One cached list per celebrity	A second copy whose wrongness is measured in seconds
Spread	Posts by author, edges by bucket	Any question that crosses a bucket, forever
Defer	Fan-out on write, below the threshold	The follower who is not in the page yet
CDN with versioned URLs	Every photo variant	Purging is eventual, so a signed link outlives the decision behind it
Bulkhead	Fan-out workers kept off the read path	Reserved capacity that sits idle on a normal day
Object-level authorization	Every hydration	A check on the read path, on every item of every page

The most useful entry here is the one we refused.

Most juniors asked to design Instagram reach for a graph database, because the words "social graph" are right there. The data kit already answered it, nearly ninety slides before this design began.

WHAT THE CARD SAID

Walk away when you have relationships but only ever join two hops.

WHAT THE FEED ACTUALLY DOES

Walks one graph hop, me to the people I follow, then a keyed lookup that is not a traversal.

WHY THE REFUSAL MATTERS MOST

A kit that only says yes is a catalog. One that says no is a tool.

Take one box out on paper, and follow what happens to the rest.

Part three set the test. Saying where each piece landed proves nothing about whether it is needed. Deleting pieces on paper is what tells you the design is finished.

THE CELEBRITY LIST CACHE

Remove it and every reader of every celebrity post reads the store directly. It stays.

THE EVENT LOG BEFORE FAN-OUT

Remove it and the write waits for the whole fan-out before returning. It stays.

THE FOLLOWER COUNT CACHE

Remove it and every display of a profile counts rows. Keep it for display, and read **B** fresh.

THE ARTIFACT

This is the whole method, and it fits on one page.

Protagonist

Antagonist

Purpose

Boundary

Environment

Constraints

Invariants

Bottleneck

Solution type

_____ wants to _____ so that _____

But _____ – and it is this big: _____

In: _____ Out: _____

physical _____ economic _____ human _____ legal _____

1 _____ 2 _____ 3 _____

_____ measured at _____

MVP • scaled • optimized • optimal • specialized

THE REVIEW

Run these six on any design, starting with your own.

- ☐ Who is the protagonist, and who are we not designing for? casting
- ☐ What is the antagonist, and what number describes it? evidence
- ☐ Which rung of the ladder is this, and does everyone agree? frame
- ☐ What breaks first at ten times the load? scale
- ☐ What happens when each dependency is slow rather than down? failure
- ☐ What can one stolen credential reach? security

YOUR TURN

Run the six review questions on the parking app, from memory.

You have both designs and the six questions. Take the smaller one: a driver, a sticker, one table, a card form.

Answer four of the six without turning back. Then turn the page.

ONE ANSWER

Four of the six, on a system you could build this month.

01	The protagonist	A driver in the rain, one hand, thirty seconds. Not the lot owner, and not the warden.
02	The antagonist	A weak signal underground, and a second tap on Pay. The number is one park charged twice.
03	Ten times the load	Nothing. Traffic was never the constraint; the manual warden was, and rung two fixed him.
04	One stolen credential	The webhook signing secret. With it a stranger marks any bay paid, which is why that endpoint checks a signature.

DEVELOP AND DELIVER

The two movements this deck skipped are the only real check on the two it taught.

Design is where the thinking lives, which is why the deck stops here. Building is where you find out whether the thinking held: the boundary you drew turns out to cut through another team, the invariant you wrote needs a lock nobody planned for, the number you estimated is wrong by ten.

So take one design you made in these pages and build the smallest version of it that actually runs. Not to ship it. To find out which of your nine fields was a guess.

BACK TO TUESDAY

482 was not blocked by the build. It was blocked by a queue.

- 01

The queue had one server

Five pull requests, one reviewer, one waking hour: a bounded pool of one. Throughput is capped, so the queue in front grows, and nothing Maya did after lunch could move any of the three.
-
- 02

Nothing inherited a deadline

The window shut at four, and nothing downstream of it carried a shorter one. The network kit's first invariant says a call inherits its deadline from the caller, and inherits a shorter one. The review never got one, so nothing said it was late until it was.
-
- 03

The one move she had, she made late

At 15:50 she stopped answering and batched the replies — admission control, from the scale kit. The spiral had run since 15:30, and those twenty minutes came out of the one hour that decided the day.

WHAT TO DO ON MONDAY

The last of these four is the one that teaches you.

STEP 01

Pick the unglamorous system

Not a famous one.
The service you were debugging on Thursday, or the pipeline nobody wants to own.

STEP 02

Cast it before you draw it

Protagonist and antagonist first, one sentence each. If you cannot name them, you do not understand it yet.

STEP 03

Fill the other seven fields

Let them argue with you. A field you cannot answer is the design question you have been avoiding.

STEP 04

Bring the page to your one-on-one

Ask the person across from you which field you got wrong. That conversation is the whole point of learning this.

HOW TO THINK ABOUT SYSTEMS

Learn the concepts and how
they connect. The technology
will change under you.

*Name the person, name the force, and the design
follows. Maya never designed anything on Tuesday, and
she met every word in Part one before she went to bed.*

Glossary

C – M

TERM	DEFINITION
CDN	Caches near readers, so bytes travel a short distance instead of an ocean.
MVP	The smallest build that puts a real answer in front of a real user.